

University of Calcutta
Mode of Examination: Offline
M.Sc. Semester – IV (Mid-Semester) Examination, 2024

Program: Master in Computer Science

Paper Code & Name: CSM 401: Elective I (Distributed & Cloud Computing)

Date: 25.03.2025

Full Marks: 30

Time: 1 Hour

Answer questions number 1 and 2 below as mentioned.

1. Answer *any FIVE*

$5 \times 2 = 10$

- a) Mention the five essential characteristics of cloud computing.
- b) Explain the measured service model of cloud computing.
- c) What is the need of cloud computing?
- d) Differentiate between single machine and virtual machine.
- e) Give an example of a private and public cloud.
- f) What is resource pooling?
- g) What are host and guest machines?

2. Answer *any FOUR*

$5 \times 4 = 20$

- a) Differentiate between Public cloud, and Private cloud.
- b. Explain the cloud computing deployment model with suitable diagram.
- c. Explain the elasticity and scalability of cloud service specialty.
- d. Explain the performance and optimization of cloud service specialty.
- e. Explain the cloud computing service model with proper diagram.

M.Sc. Semester – IV (Mid-Semester) Examination, 2025
Paper Code & Name: CSM 401: Elective I (Distributed & Cloud Computing)

Date: 13.05.2025

Full Marks: 30

Time: 1 Hour

CT2

Answer questions number 1 and 2 below as mentioned.

1. Answer *any FIVE*

$$5 \times 2 = 10$$

- a) What is data virtualization?
- b) Explain access virtualization.
- c) What is Kubernetes?
- d) Explain the benefits of Kubernetes.
- e) What is a Container?
- f) Distinguish between Container and Hypervisor.

2. Answer *any FOUR*

$$5 \times 4 = 20$$

- a. Distinguish between access virtualization from application virtualization.
- b. Explain the process of Containerization with a suitable diagram.
- c. Write the benefits and limitations of Virtualization.
- d. What is memory virtualization and explain different types of memory virtualization.
- e. Explain different types of hardware virtualization.